SECTION 1 – INTRODUCTION

1-1 HOW THE PELVIC PHASED ARRAY COIL OPERATES

The Pelvic Phased Array Coil is a four element receive only antenna array which receives the RF signal from the body and sends it to the preamplifier. It is used to image the pelvic region. The small individual antenna loop elements (4 in total) provide excellent SNR while the “multicoil” concept provides a large field of view (FOV). The overall length of the antenna portion of the array is 20 cm. The PELVIC_TOP utilizes antenna elements 1 and 2. The PELVIC_BOTTOM utilizes antenna elements 3 and 4. The Pelvic Coil is tuned to the MR frequency for proton, 21.29 MHz.

The Coil consists of three major assemblies: (See Illustration 1–1.)

- Pelvic – Top Circuit Board (1) 46–321008G1
- Pelvic – Bottom Circuit Board (1) 46–321010G1
- Phase Shift Board (1) 46–321012G1

The 50 ohm phase shift network on the Phase Shift Board ensures a multiple of a half wavelength from the 30 pin connector to the coil input. The coils themselves are made up of an input blocking circuit, the antenna loop inductance and loss and the distributed capacitance. The geometry of the coil is chosen to optimize performance. This geometry dictates the loop inductance and loaded coil losses. The distributed capacitors are chosen to resonate with the loop inductance. The input capacitor of the coil (Cin) is chosen such that the loaded coil looks like 50 ohms at the preamp input. The air wound blocking inductor is dictated by the value of Cin such that the blocking circuit resonates at 21.29 MHz.

Each individual loop element is capacitively loaded with lumped element chip capacitors to reduce patient loading effects. The total capacitance value of all the series chip capacitors combined is chosen in conjunction with the input capacitor C_{in} to allow for a matched impedance ($Z_{in}=50\Omega$) at the antenna input. No tuning capacitor is needed because all capacitors are tight tolerance matched pairs (¿ 1%).

The input circuitry for each antenna element in the Pelvic array serves two functions. The first function is to match the input impedance of the individual antenna loop elements to the desired value of 50Ω. This is accomplished through the use of a simple *ell* matching network consisting of a series inductor L_{in} and shunt capacitor C_{in}.

The second function performed by the input circuitry on the Pelvic array is that of transmit blocking. This function is actually performed by both the input matching network as well as the transmit blocking network, thereby providing twice the current blocking of a single network at a given bias level.

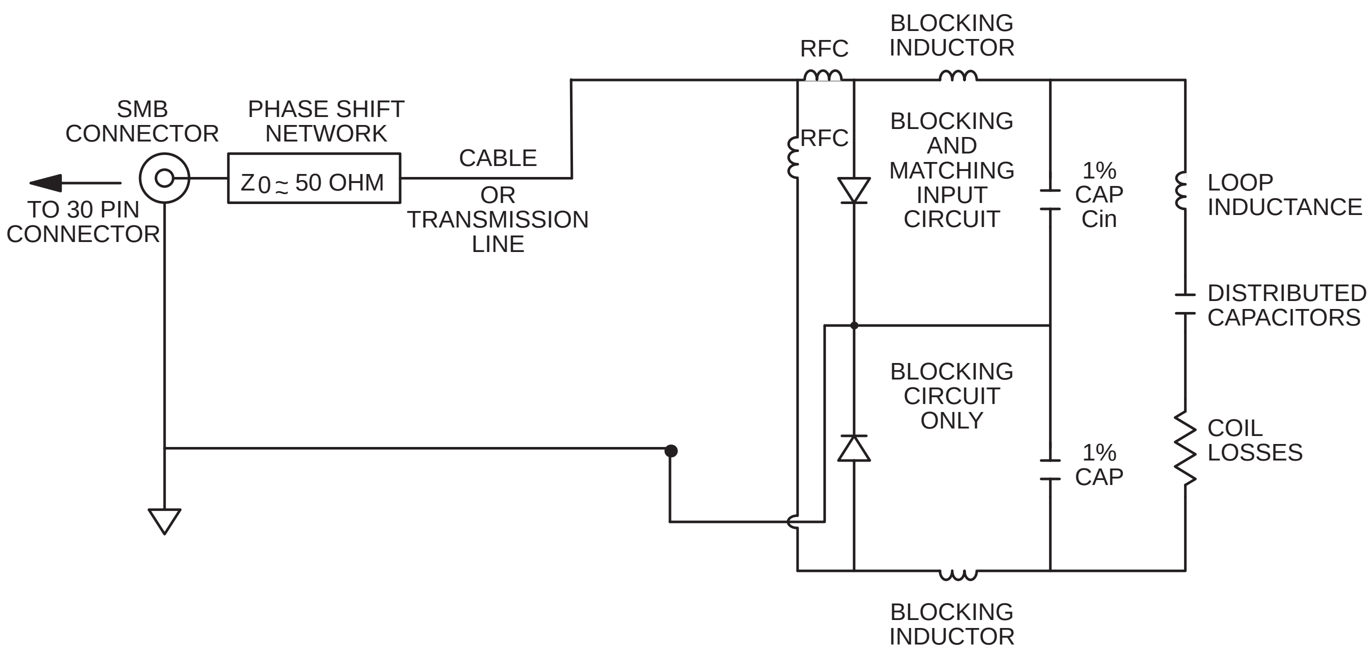
The transmit blocking network (and input matching network) are simple high impedance parallel resonant LC circuits tuned to resonate at f_o=21.29 MHz. The resonant circuits are closed (or switched into the antenna loop) during transmit by forward biasing the PIN diode. This bias current is supplied from the Multicoil Switch Driver board and is split equally between the two diodes (one for each network).

1–1 HOW THE PELVIC PHASED ARRAY COIL OPERATES (continued)

During transmit all PIN diodes are biased on. With the diode turned on, the LC blocking network resonates to form a high impedance at the input of the coil. This minimizes currents in the coil loop which eliminates the magnetic field produced by the coil. If a magnetic field were produced by the coil, the transmit field generated by the body coil would be affected and homogeneity would be lost.

During receive all PIN diodes are biased off. Magnetic coupling from nearby coils is reduced by using the low input impedance of the preamp. The electrical length of the system from the preamp to coil input transforms this low impedance (ie: short circuit) across the diode and the LC network once again blocks coil loop currents. This ensures that the individual signals picked up by the four separate preamps are not contaminated. Coupling between adjacent coils is further reduced by overlapping them.

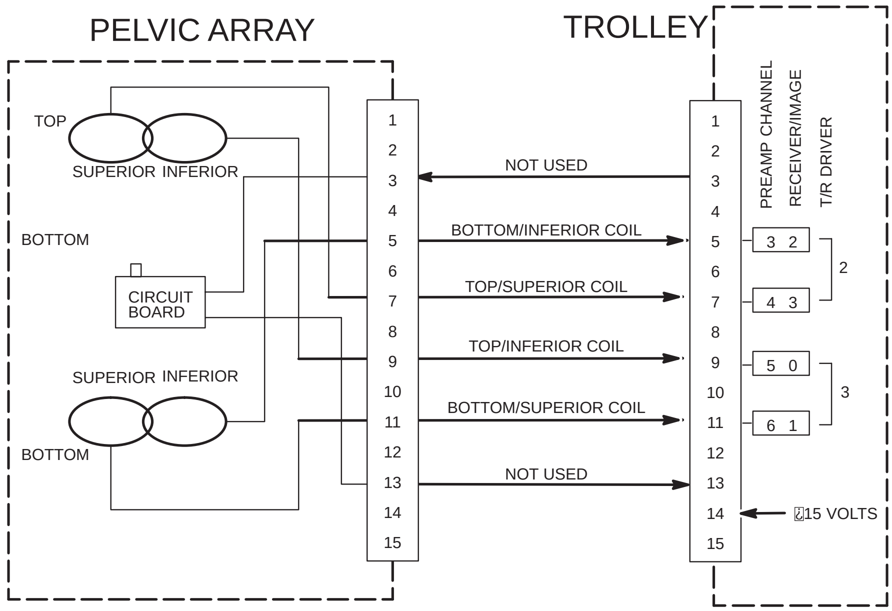
At the factory, the individual loop elements are tested as circuit board pairs to ensure proper input impedance. Since the antenna elements are contained on two circuit boards, neither of these boards can be replaced separately (these circuit boards include the Pelvic – Top Circuit Board and Pelvic – Bottom Circuit Board). Only the Phase Shift Board (46–321012G1) can be removed and replaced separately.



BLOCK DIAGRAM FOR PELVIC PHASED ARRAY COIL
ILLUSTRATION 1–1

1–1 HOW THE PELVIC PHASED ARRAY COIL OPERATES (continued)

IMAGE NUMBER	PELVIC ARRAY ONLY
1	TOP/INFERIOR COIL
2	BOTTOM/SUPERIOR COIL
3	BOTTOM/INFERIOR COIL
4	TOP/SUPERIOR COIL



INTERCONNECTIONS FOR PELVIC PHASED ARRAY COIL
ILLUSTRATION 1–2

1-2 COMPATIBILITY

The Pelvic Phased Array Coil is compatible with the following hardware configurations:

- Signa Advantage™ 0.5T (0.5T only)

1-3 RELATED DOCUMENTS

- *Direction 15404, Signa Advantage™ 1.5T & 0.5T Renewal Parts*

1-4 ORGANIZATION OF THIS DOCUMENT

This manual is divided into the following sections:

Section 1, INTRODUCTION, describes how the Pelvic Phased Array Coil operates and when and where the Pelvic Phased Array Coil can be used.

Section 2, SET UP AND CALIBRATION, describes installation procedures.

Section 3, FUNCTIONAL CHECKS, describes the normal power-up sequence.

Section 4, REPLACEMENT/MAINTENANCE, describes field maintenance procedures.

Section 5, RENEWAL PARTS, lists field replaceable parts.

Appendix A, assembly drawings and schematics

1-5 ENVIRONMENTAL REQUIREMENTS

Operate and store the Pelvic Phased Array Coil in the Scanner Room.

1-6 SPECIFICATIONS

Parameter	Value
Coil design	Four element receive only antenna array for human pelvic imaging
Tuned Frequency	21.29 MHz (factory set)
Optimum field of view	Group of four antenna loops providing 20cm FOV (longitudinally)
Operating and storage environment	Operate and store the Pelvic Phased Array Coil in the Scanner Room

SECTION 2 – SET UP AND CALIBRATION

2-1 CHECKING THE SHIPPING LIST

Table 2-1 lists the M1687AC Signa Advantage 0.5T Pelvic Phased Array Coil parts. Check that all parts have been shipped.

TABLE 2-1
PELVIC PHASED ARRAY COIL SHIPPING LIST

QTY.	ITEM	PART NO.
1	Pelvic Phased Array Coil and Single Cable	46–320972G1
1	Strap	46–306594P1
1	Positioner and Spherical Phantoms	46–317626G1
1	Service Manual	46–305149

2-2 INSTALLING THE PELVIC PHASED ARRAY COIL

At the Boot Terminal, install a new soft key. This key will be used by the operator to select PELVIC (elements 1–4) imaging. Name the key “PELVIC”. See Illustration 2–1.

Refer to the Signa Advantage or Signa Advantage 1.5T & 0.5T System Manuals for information on installing soft keys (use Coil/Phased Array default values in Table 2–2 and 2–3).

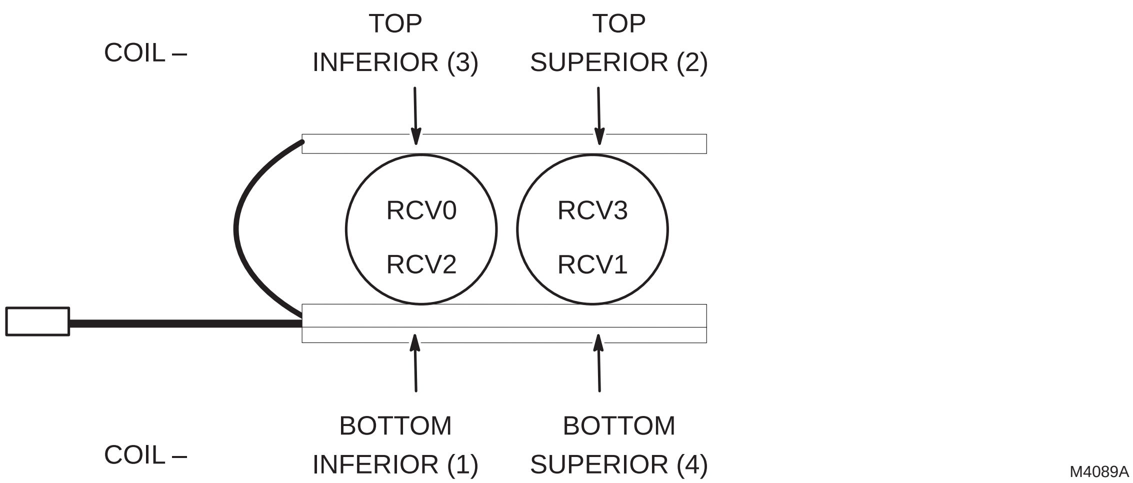
TABLE 2-2
COIL VALUES

SYS-TEM	COIL NAME	COIL TYPE	EXTREM-ITY COIL	CABLE LOSS	COIL LOSS	RECON SCALE FACTOR	XMIT COIL TYPE	XMIT ATTN *(Rel 5.2 only)*	MULTI-COIL
0.5T	PELVIC	surface	**no**	**1.05**	**0.313**	Head Coil Recon Scale Factor x 1.00	**quad**	**0**	**yes**

TABLE 2-3
PHASED ARRAY VALUES

MULTI–COIL NAME	NUMBER OF RECEIVERS	START RECEIVER	STOP RECEIVER	PORT ENABLE MASK	ERROR ENABLE MASK
PELVIC	**4**	**0**	**3**	**6**	**6**

2–2 INSTALLING THE PELVIC PHASED ARRAY COIL (continued)



PELVIC PHASED ARRAY COIL TO TPS RECEIVER CORRESPONDENCE
ILLUSTRATION 2–1

2-3 FUNCTIONAL CHECKS

1. Perform a Body coil scan SNR verification. Refer to Section 3–1, BODY COIL SNR VERIFICATION.

2. Perform a Pelvic Phased Array Coil SNR Verification. Refer to Section 3–2, PELVIC PHASED ARRAY COIL SNR VERIFICATION.

3. Perform Spike Noise Analysis for the Pelvic Phased Array Coil. Refer to *Direction 15300, Signa Advantage™ System,* Performance Checks tab, Section 2–1, SPIKE NOISE or *Direction 15400, Signa Advantage™ 1.5T & 0.5T System,* Performance Checks tab, Section 3, SPIKE NOISE.

2-4 PERIODIC QUALITY ASSURANCE CHECK

On a periodic basis, such as during planned maintenance, perform the quality assurance check outlined below to ensure that the Pelvic Phased Array Coil is operating properly with no appreciable degradation of image quality.

1. Check the external cable for cracks or breaks once a week. Refer to Section 4–6, CHECKING THE CABLES.
2. Perform a Pelvic Phased Array Coil SNR Verification. Refer to Section 3–2, PELVIC PHASED ARRAY COIL SNR VERIFICATION.
3. Record the date and value calculated in Section 3–3, SNR IMAGE ANALYSIS in column 2 under “SNR Data QA Check” of the Data Table as is instructed.
4. As is instructed in the Data Table, divide the SNR value obtained in the periodic QA check (recorded in column 2) by the original SNR value (recorded in column 3) and record in column 4 of the Data Table.
5. If this ratio is not greater than 85%, then there may be a problem in the Multicoil system. Contact your local GE Service Representative.

SECTION 3 – FUNCTIONAL CHECKS

3-1 BODY COIL SNR VERIFICATION

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to ”TLT PROCEDURE” calibration procedure located in Direction 15491, *Signa Advantage 1.5T & 0.5T System Troubleshooting Guide*.

Phantom Required (shipped with each corresponding Pelvic Phased Array Coil)

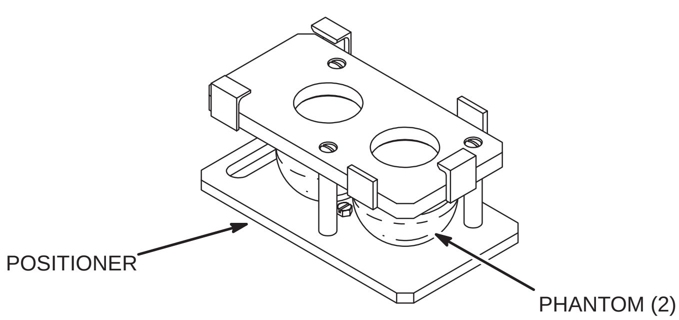
- Pelvic Phased Array Coil Positioner with phantoms (Pelvic Phantom/Positioner), 46–317626G1

Setup Procedure

CAUTION

The Quad Head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Head Coil T/R Network.

1. Remove Quad Head Coil (if present) from cradle.
2. Select **[New Exam]** (5.x) to allow a new Landmark to be set.
3. Position the Pelvic Phantom/Positioner with phantoms in the center of the cradle. Landmark the center of the positioner and advance the positioner with phantoms to isocenter using the **[ADVANCE TO SCAN]** button. See Illustration 3–1.



PELVIC PHANTOM/POSITIONER LANDMARK SETUP
ILLUSTRATION 3–1

3–1 BODY COIL SNR VERIFICATION (continued)

4. Setup Scan Prescription as shown in Table 3–1 (5.x).

TABLE 3–1
BODY COIL SNR – SCAN PROTOCOL (5.x)

SCAN PRESCRIPTION (5.x)	
MAIN MENU	**[New Exam]**
PATIENT/EXAM INFORMATION	
Id:	**geservice**
Name:	**body snr**
Patient Weight	**300**
	[Patient Position]
PATIENT POSITION	
Patient Entry	**[Head First]**
Patient Position	**[Supine]**
Axial/Sag. Landmark	**[Sternal Notch]**
Coil Type	**[Body Coil]**
Scan Plane	**[Axial]**
	[Image Params]
IMAGING PARAMETERS	
Image Mode	**[2D]**
(K *SAR must be "On"* K)	**[Monitor SAR]**
Pulse Sequence	**[Spin Echo]**
Imaging Options	**[None]**
or enter PSD Filename	
	[Scan Timing] or [Next Screen]
SCAN TIMING	
Number of Echoes	**[1]**
Echo Time (TE)	**[20 msec]**
Rep Time (TR)	**[2000 msec]**
	[Scan Set–Up] *(Release 5.3)*
	[Scanning Range] *(Release 5.2)*
SCAN SET–UP *(Release 5.3)*	
Auto CF	**[Peak]**
Receive Bandwidth (+/–)	*for 0.5T ONLY* **[16kHz]**
	[Scanning Range]
SCANNING RANGE	
Field of View	**[48 cm]**
Scan Thickness	**[3 mm]**
Interscan Spacing	**[Other] 0**
Start Loc (I/S):	**0**
End Loc (I/S):	**0**
No. of Scan Locations:	1
FOV Center (L/R):	0 (P/A): 0
	[#] **[Acq Time]**
ACQUISITION TIME	
Acq. Matrix (freq.)	**[256]**
Acq. Matrix (phase)	**[256]**
Frequency Direction	**[A/P]**
Phase FOV	default *(Release 5.3)*
Imaging Time	**[1 NEX 8:58]**
Contrast	**[No]**
Table Delta:	0 mm
	[Scan Ops] *(Release 5.3)*
	[Scan Set–Up] *(Release 5.2)*
SCAN SET–UP *(Release 5.2)*	
Auto CF	**[Peak]**
Receive Bandwidth (+/–)	*for 0.5T ONLY* **[16kHz]**
	[Scan Ops]
SCAN OPERATIONS	

5. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90 degree and 180 degree pulses.
6. Select **[Scan]**. Observe the resulting images. Ensure that there are no artifacts of any sort in the resulting image. Record the Exam number and Series number for SNR Calculations.
7. Select **[Scan]** again. This second image will be used for determination of Body Coil mode SNR.
8. Select **[Cancel]**. Refer to Section 3–3 for SNR image analysis.

3-2 PELVIC PHASED ARRAY COIL SNR VERIFICATION

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to ”TLT PROCEDURE” calibration procedure located in Direction 15491, *Signa Advantage 1.5T & 0.5T System Troubleshooting Guide*.

Phantom Required (shipped with each corresponding Phased Array Coil)

- Pelvic Phased Array Coil Positioner with phantoms (Pelvic Phantom/Positioner), 46–317605G1

Setup Procedure

1. Select **[New Exam]** (5.x) to allow a new Landmark to be set.

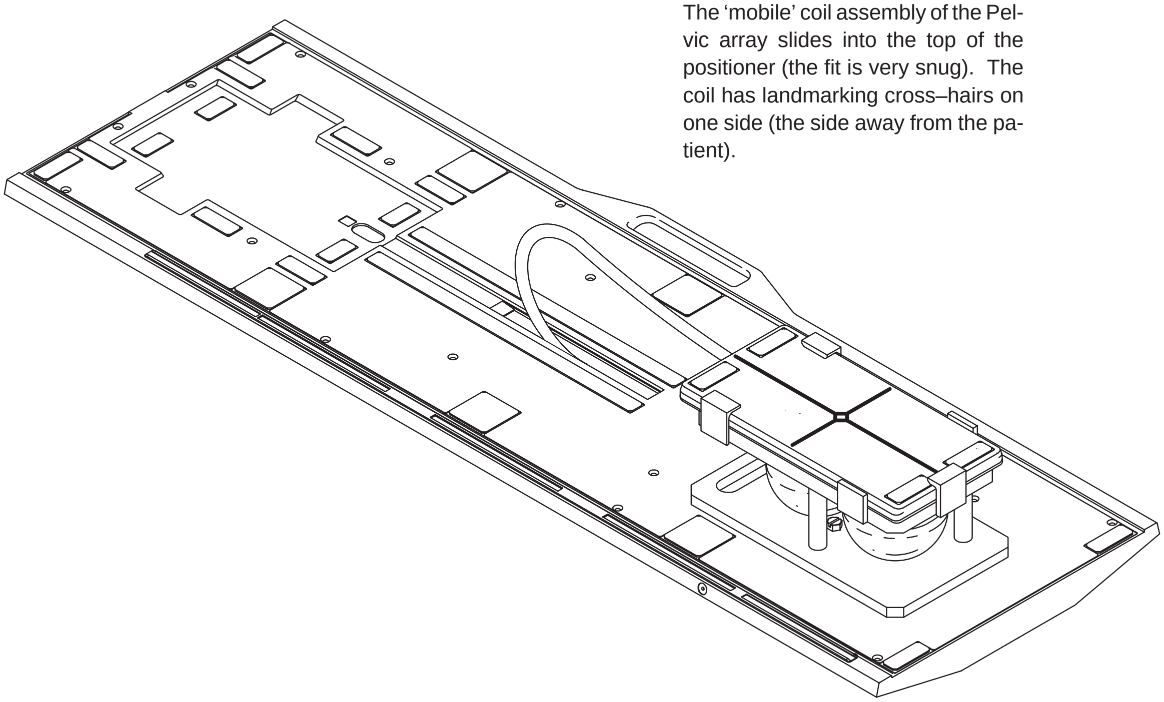
CAUTION

The Quad head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Head Coil T/R Network.

2. Remove Quad Head Coil (if present) from cradle.
3. Place Pelvic Phased Array Coil to be tested onto table.
4. Connect Pelvic Phased Array Coil connector to its mating connector in the Carriage Assembly.
5. Place the phantom/positioner down on the Pelvic Coil. Ensure the two pegs on the bottom of the positioner mate with two positioning holes on the top cover of the Pelvic Coil Array. See Illustration 3–2.

3–2 PELVIC PHASED ARRAY COIL SNR VERIFICATION (continued)

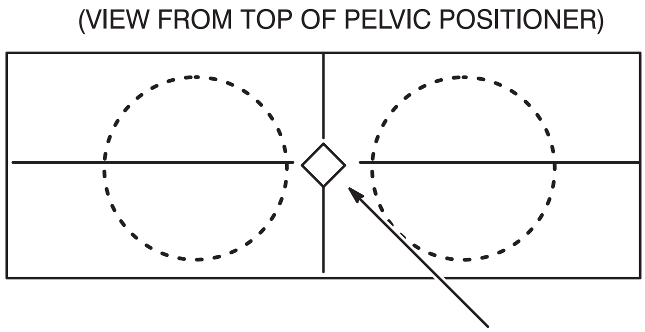
The ‘mobile’ coil assembly of the Pelvic array slides into the top of the positioner (the fit is very snug). The coil has landmarking cross–hairs on one side (the side away from the patient).



PELVIC PHANTOM/POSITIONER SETUP
ILLUSTRATION 3–2

3–2 PELVIC PHASED ARRAY COIL SNR VERIFICATION (continued)

6. Position the Pelvic Phased Array Coil and Positioner with phantoms in the center of the cradle. Landmark for Pelvic MID and advance the positioner with phantoms to isocenter using the **[ADVANCE TO SCAN]** button. See Illustration 3–3.

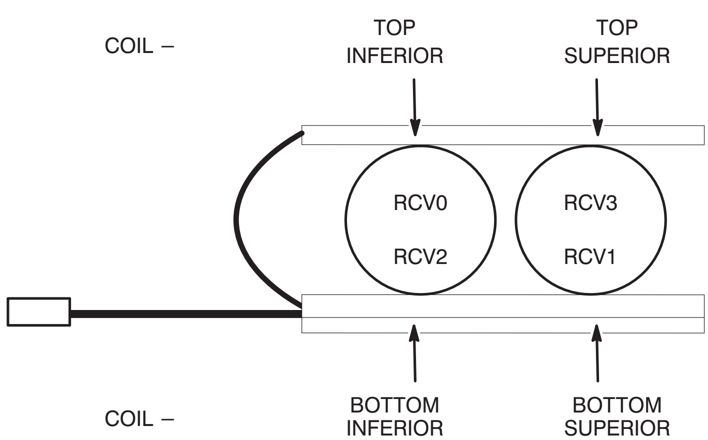


Note: Testing the Pelvic Coil requires one scan. Landmark using the cross–hairs on the Pelvic array (they will be apparent once you place the coil in the positioner).

PELVIC PHANTOM/POSITIONER LANDMARK SETUP (PELVIC TOP)
ILLUSTRATION 3–3

Phased Array Coil Receiver Paths

The Phased Array Coils have multiple individual antenna loop elements inside the assembly; the TPS chassis has multiple receivers (RCV0–RCV3). See Illustration 3–4.



PELVIC PHASED ARRAY COIL TO TPS RECEIVER CORRESPONDENCE
ILLUSTRATION 3–4

3–2 PELVIC PHASED ARRAY COIL SNR VERIFICATION (continued)

7. Setup Scan Prescription as shown in Table 3–2 (5.x).

TABLE 3–2
PELVIC PHASED ARRAY COIL SNR – SCAN PROTOCOL (5.x)

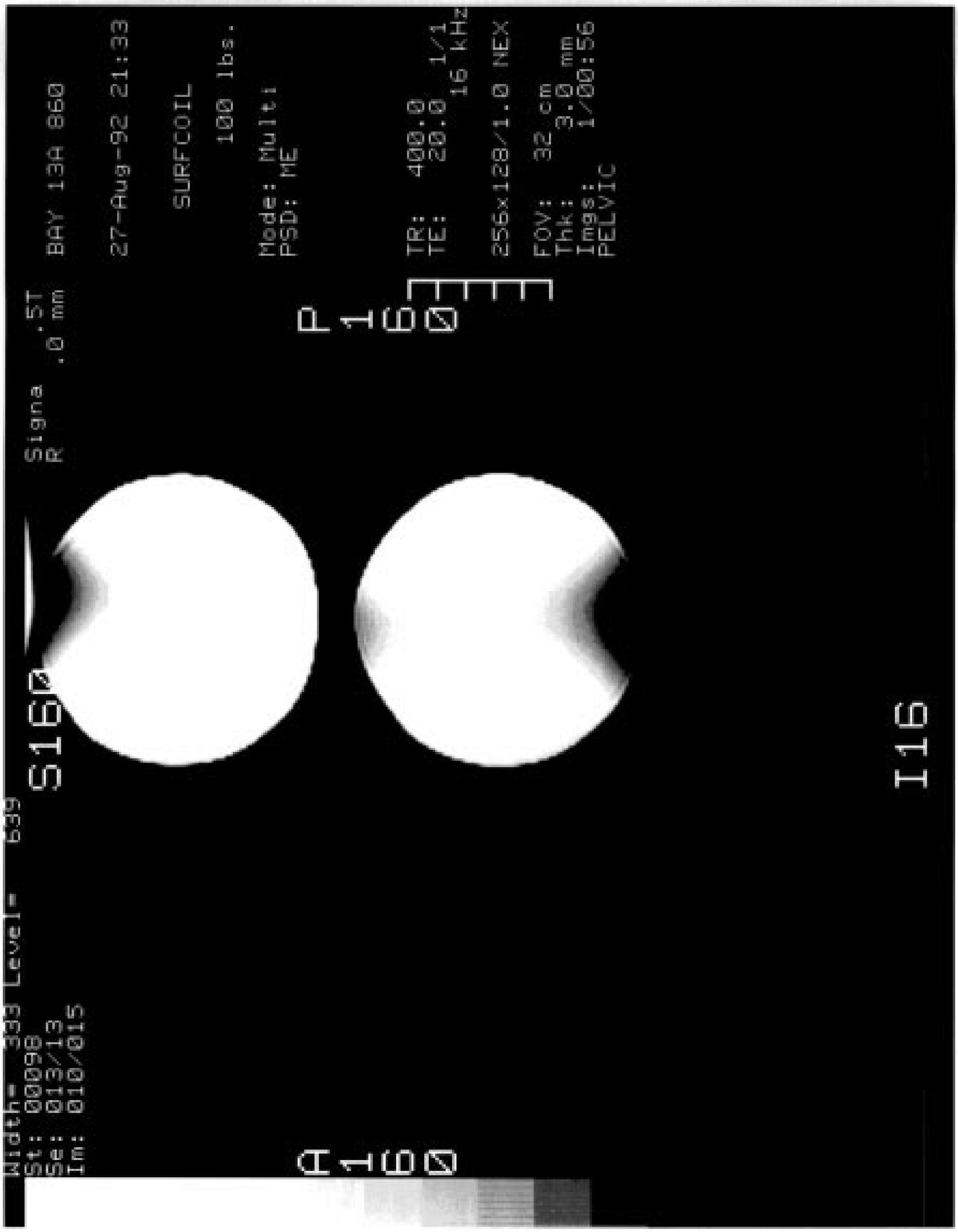
SCAN PRESCRIPTION (5.x)			
MAIN MENU	**[New Exam]**	SCAN SET–UP *(Release 5.3)*	
PATIENT/EXAM INFORMATION		Auto CF	**[Peak]**
Id:	**geservice**	Receive Bandwidth (+/–)	*for 0.5T ONLY* **[16kHz]**
Name:	**pelvic snr**		**[Scanning Range]**
Patient Weight	**300**	SCANNING RANGE	
	[Patient Position]	Field of View	**[48 cm]**
PATIENT POSITION		Scan Thickness	**[3 mm]**
Patient Entry	**[Head First]**	Interscan Spacing	**[Other] 0**
Patient Position	**[Supine]**	Start Loc (I/S):	**0**
Axial/Sag. Landmark	**[Sternal Notch]**	End Loc (I/S):	**0**
Coil Type	**[Other Coils] [PELVIC]**	No. of Scan Locations:	1
Scan Plane	**[Axial]**	FOV Center (L/R):	0 (P/A): 0
	[Image Params]		[#] **[Acq Time]**
IMAGING PARAMETERS		ACQUISITION TIME	
Image Mode	**[2D]**	Acq. Matrix (freq.)	**[256]**
(K *SAR must be "On"* K)	**[Monitor SAR]**	Acq. Matrix (phase)	**[256]**
Pulse Sequence	**[Spin Echo]**	Frequency Direction	**[A/P]**
Imaging Options	**[None]**	Phase FOV	default *(Release 5.3)*
or enter PSD Filename		Imaging Time	**[1 NEX 8:58]**
	[Scan Timing] or [Next Screen]	Contrast	**[No]**
SCAN TIMING		Table Delta:	0 mm
Number of Echoes	**[1]**		**[Scan Ops]** *(Release 5.3)*
Echo Time (TE)	**[20 msec]**		**[Scan Set–Up]** *(Release 5.2)*
Rep Time (TR)	**[2000 msec]**	SCAN SET–UP *(Release 5.2)*	
	[Scan Set–Up] *(Release 5.3)*	Auto CF	**[Peak]**
	[Scanning Range] *(Release 5.2)*	Receive Bandwidth (+/–)	*for 0.5T ONLY* **[16kHz]**
			[Scan Ops]
		SCAN OPERATIONS	

8. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90 degree and 180 degree pulses.

9. Select **[Scan].** Observe the resulting image of the spheres. See Illustration 3–5 (normal image). Ensure that there are no artifacts of any sort in the sphere images. Record the Exam number and Series number for SNR Calculations.

10. Select **[Scan]** again. This second image of the spheres will be used for determination of Pelvic mode SNR.

3–2 PELVIC PHASED ARRAY COIL SNR VERIFICATION (continued)



PELVIC PHASED ARRAY COIL IMAGE
ILLUSTRATION 3–5

3-3 SNR IMAGE ANALYSIS

Description

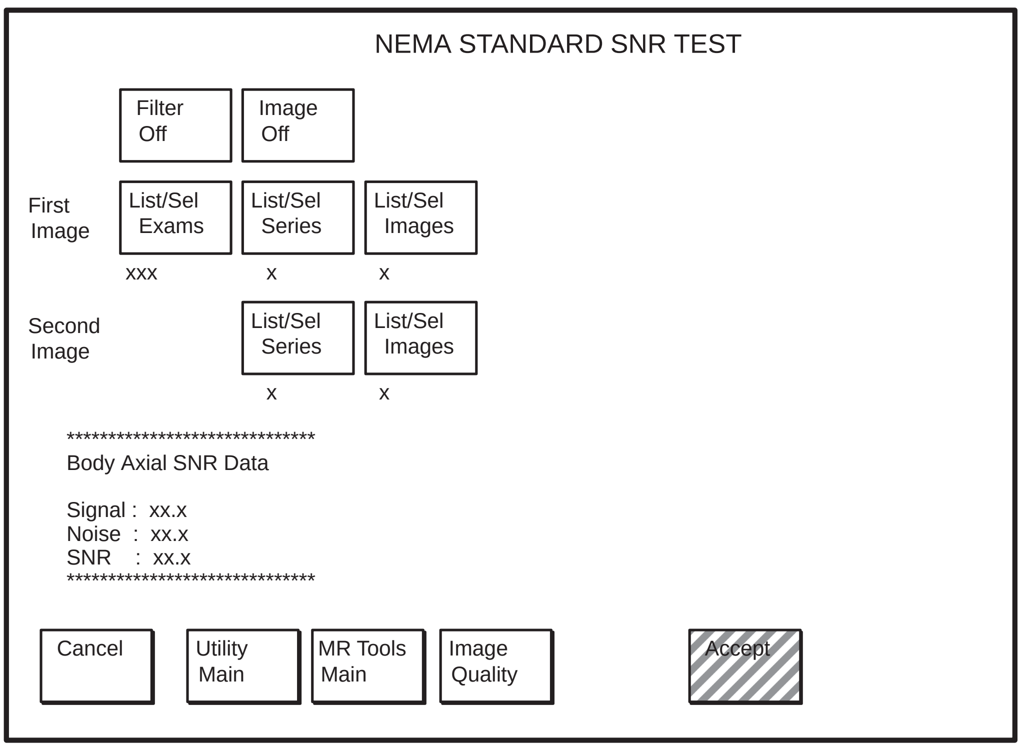
The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center of the image for positioning the ROI. A difference image is created by subtracting the second image from the first and the same ROI is used to calculate noise from the subtracted image. The signal value, noise value, and signal to noise ratio are reported. There is an option to save the difference image with the results annotated.

Procedure

1. Touch **[UTILITIES], [MR Tools], [Image Quality],** then **[SNR Test]**. The SNR Test screen is displayed; see Illustration 3–6.
2. Enter image exam, series, and image numbers. If exam, series, or image numbers are not known, select **[List Exams], [List Series],** or **[List Images]** to display list to choose from.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.



Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN
ILLUSTRATION 3–6

3–3 SNR IMAGE ANALYSIS (continued)

3. If high pass filtering is desired to be performed on data, touch **[Filter Off]** which will highlight and change to **[Filter On]**.
4. If the difference image annotated with data is to be created, touch **[Image Off]** which will highlight and change to **[Image On]**.
5. Touch **[Accept]** to begin analysis. The final values are displayed on the touch screen, see Illustration 3–6.
6. Touch **[Continue]** then select the next exam and repeat the above analysis for each image pair.
7. Record the date and value calculated in the appropriate column under “SNR Data QA Check” of the Data Table on page DT–1 (following Section 5, RENEWAL PARTS) as is instructed.

3-4 CHECKING THE PROTECTION PIN DIODES WITH DIGITAL MULTIMETER (DMM)

Note

Due to the fact that there are two PIN diodes in parallel for each antenna element, this procedure will not indicate if **one** of the PIN diodes is defective, only if **both** are defective.

1. Select the DIODE TEST function on the Digital Multimeter (DMM).
2. Connect the NEGATIVE lead of the DMM to the external cable connector row A or B, Pins 1, 3, 5, and 7. Connect the POSITIVE lead to the external cable connector row A or B, Pin 14 (ground). Refer to Table 3–3.
3. A reading of 0.400 to 0.900 should be observed on the DMM.
4. If a reading below 0.400 is observed in either direction, either the output cable is shorted or bad PIN diode.
5. If a reading above 0.900 is observed in step 2, both PIN diodes are defective.
6. Connect the NEGATIVE lead of the DMM to the external cable connector row A or B, Pin 14 (ground). Connect the POSITIVE lead to the external cable connector row A or B, Pins 1, 3, 5, and 7. Refer to Table 3–3.
7. A reading of INFINITY should be observed on the DMM.
8. If a reading of INFINITY is observed in both directions, either the output cable is open or **both** PIN diodes are open.
9. If any of the above conditions fails, replace the coil. **The Pin diodes are not FRU's.**

TABLE 3–3
DIODE TEST CONNECTIONS

	POSITIVE LEAD CONNECTION		NEGATIVE LEAD CONNECTION	
COIL NUMBER	FOR STEP 2	FOR STEP 4	FOR STEP 2	FOR STEP 4
1	A14 or B14	A1 or B1	A1 or B1	A14 or B14
2	A14 or B14	A3 or B3	A3 or B3	A14 or B14
3	A14 or B14	A5 or B5	A5 or B5	A14 or B14
4	A14 or B14	A7 or B7	A7 or B7	A14 or B14

3-5 CHECKING THE EXTERNAL CABLE

1. Select the DIODE TEST function on the Digital Multimeter (DMM).
2. Connect the NEGATIVE lead of the DMM to the external cable connector row A or B, Pins 1, 3, 5, and 7. Connect the POSITIVE lead to the external cable connector row A or B, Pin 14 (ground). Refer to Table 3–3.
3. Flex the external cable, especially near the connectors and the strain relief, and observe that a reading of 0.400 to 0.900 should remain on the DMM, with no instability or fluctuations.
4. Connect the NEGATIVE lead of the DMM to the external cable connector row A or B, Pin 14 (ground). Connect the POSITIVE lead to the external cable connector row A or B, Pins 1, 3, 5, and 7. Refer to Table 3–3.
5. Flex the external cable, especially near the connectors and the strain relief, and observe that a reading of INFINITY should remain on the DMM, with no instability or fluctuations.
6. If the cable fails any of the above tests, it should be replaced.

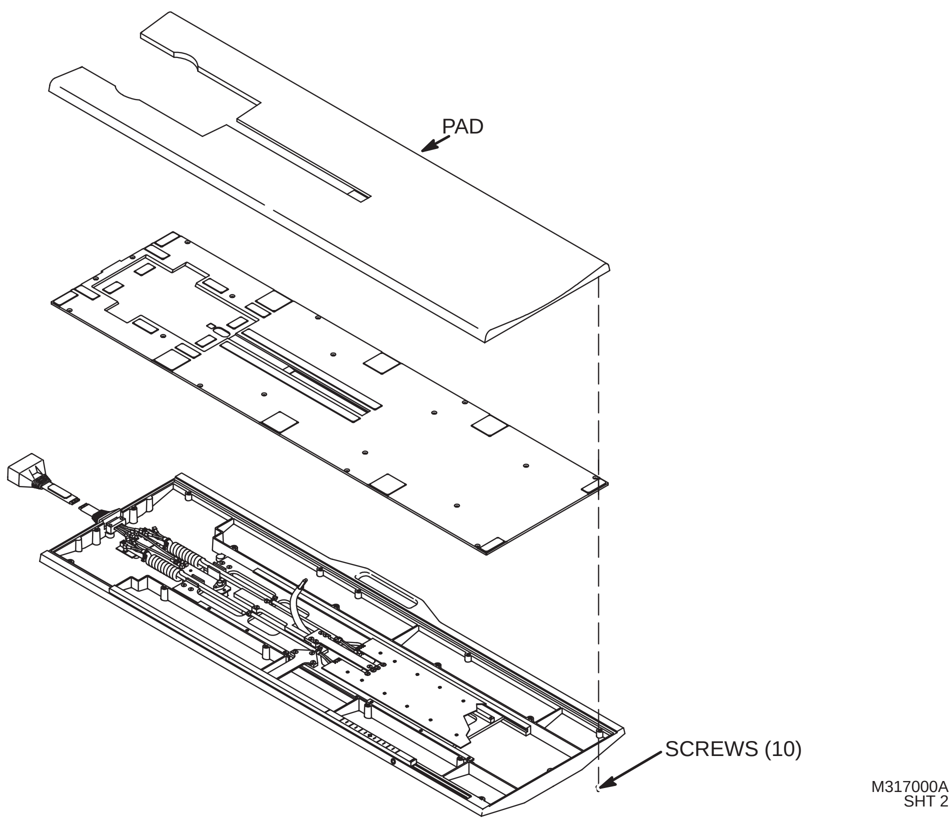
SECTION 4 – REPLACEMENT/MAINTENANCE

Note

All antenna loop elements are tuned at the factory to provide the proper input impedance. Since antenna elements 1 and 2 are connected to elements 3 and 4, the top and bottom Pelvic Array circuit boards are field replaceable as a unit. These circuits must remain together as a matched set.

4-1 DISASSEMBLY/ASSEMBLY OF PELVIC PHASED ARRAY COIL

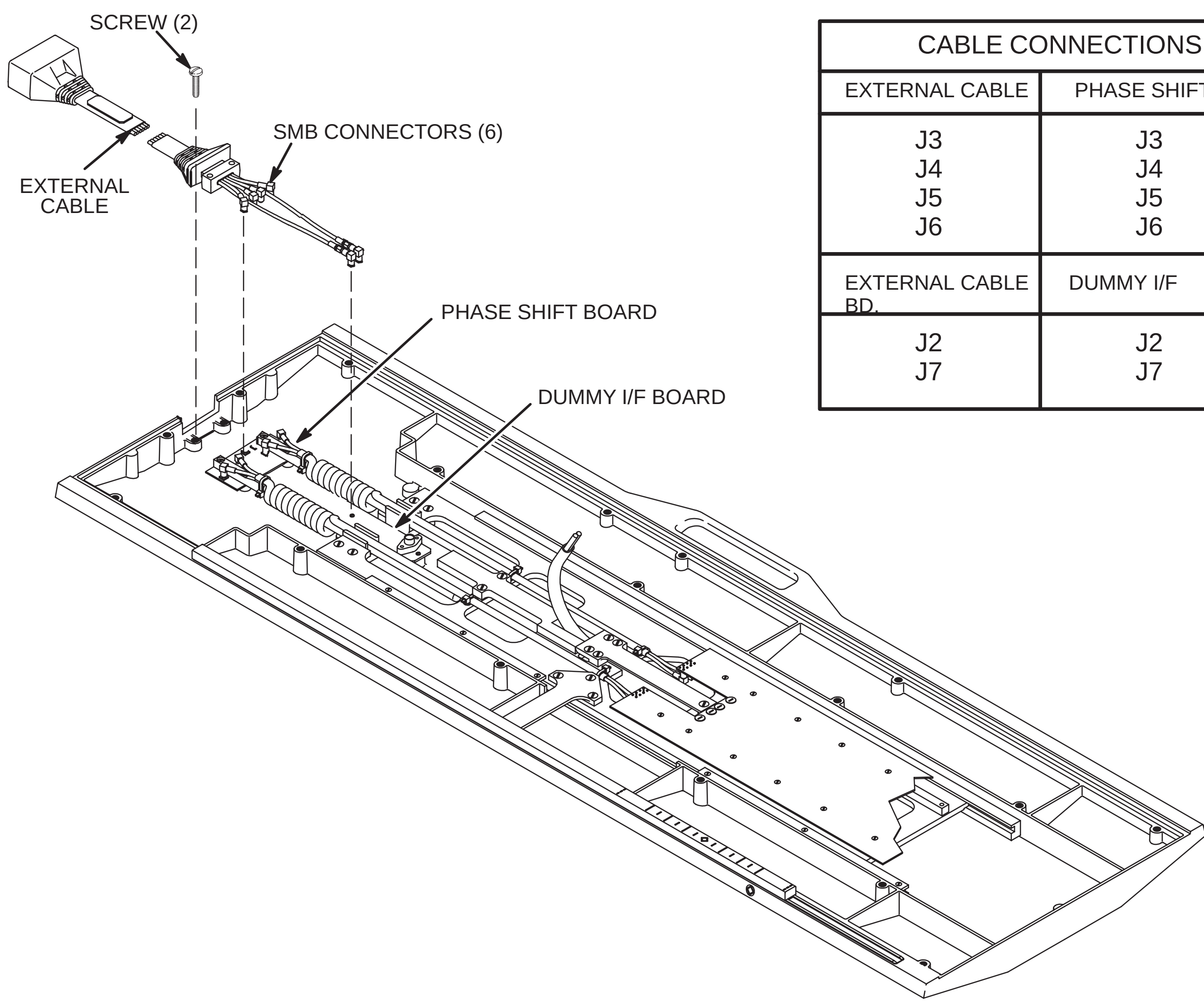
1. Remove pad from top side of Pelvic Coil assembly. See Illustration 4–1.
2. Remove the ten screws holding the top and bottom housing together.
3. Separate the two housings from each other.
4. To reassemble, put the two housings together and tighten the ten screws.
5. Attach pad to top side of Pelvic Coil assembly.



DISASSEMBLY/ASSEMBLY OF PELVIC PHASED ARRAY COIL
ILLUSTRATION 4–1

4-2 REPLACING THE EXTERNAL CABLE AND INTERFACE CIRCUIT BOARD

1. Disassemble the top and bottom housings. Refer to Section 4–1.
2. Remove the two screws holding the external cable in place.
3. Disconnect the six SMB connectors on the external cable from the Phase Shift Board (four connectors) and Dummy Interface Board (two connectors). See Illustration 4–2.
4. Remove the bad external cable.
5. Install the new external cable in place. See Illustration 4–2.
6. Install the screws in place.
7. Reconnect the six SMB connectors to the Phase Shift Board and Dummy Interface Board in the proper order.
8. Assemble the top and bottom housings together. Refer to Section 4–1.



CABLE CONNECTIONS	
EXTERNAL CABLE	PHASE SHIFT BD.
J3 J4 J5 J6	J3 J4 J5 J6
EXTERNAL CABLE BD.	DUMMY I/F
J2 J7	J2 J7

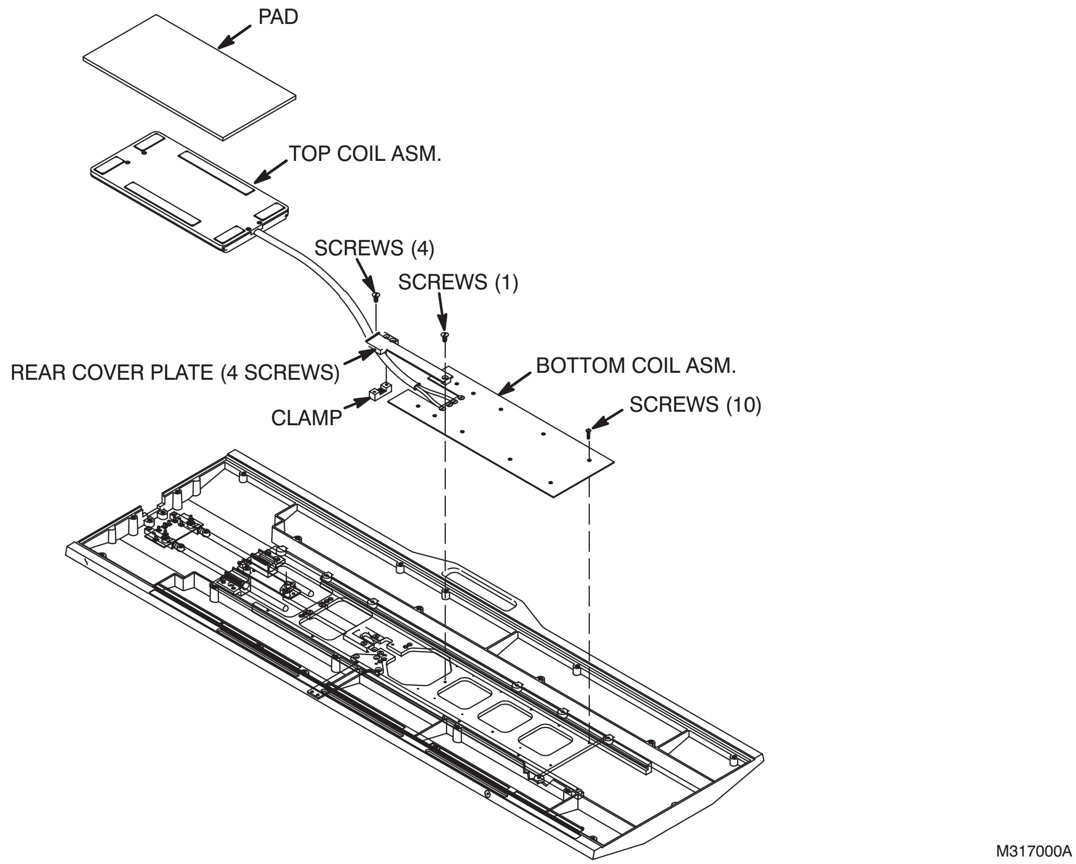
REPLACEMENT OF EXTERNAL CABLE
ILLUSTRATION 4–2

4-3 REPLACING THE TOP AND BOTTOM PELVIC ARRAY CIRCUIT BOARDS

Note

Cable lengths are tuned. Therefore, internal cable replacement must involve the correct replacement cable of appropriate length.

1. Disassemble the top and bottom housings. Refer to Section 4–1.
2. Remove the rear cover plate and the clamp holding the internal cable in place. See Illustration 4–3.
3. Disconnect the four coil cord cables from the Bottom Coil Assembly.
4. Remove the ten screws from Bottom Coil Assembly.
5. Remove the Top and Bottom Coil Assembly.
6. Install the new Top and Bottom Coil Assembly. See Illustration 4–3.
7. Connect the four coil cord cables from the Bottom Coil Assembly.
8. Install the ten screws to Bottom Coil Assembly.
9. Assemble the top and bottom housings together. Refer to Section 4–1.



REPLACEMENT OF TOP AND BOTTOM PELVIC ARRAY CIRCUIT BOARDS
ILLUSTRATION 4–3

4-4 REPLACING THE PROTECTION PIN DIODES

CAUTION

As coil tuning and decoupling tuning may be affected by PIN diode replacement, field replacement of the PIN diodes shall not be performed.

4-5 REPLACING THE MECHANICAL HARDWARE

1. Disassemble the top and bottom housings, if necessary. Refer to Section 4–1.
2. Disconnect and remove the bad part. Refer to Section 5, Renewal Parts, for part numbers.
3. Install the new part in place.
4. Assemble the top and bottom housings together. Refer to Section 4–1.

4-6 CHECKING THE CABLES

Check the external cable for cracks or breaks once each week. Replace the external cable if any damage or wear is found. Refer to Section 4–2, REPLACING THE EXTERNAL CABLE, for external cable replacement procedure.

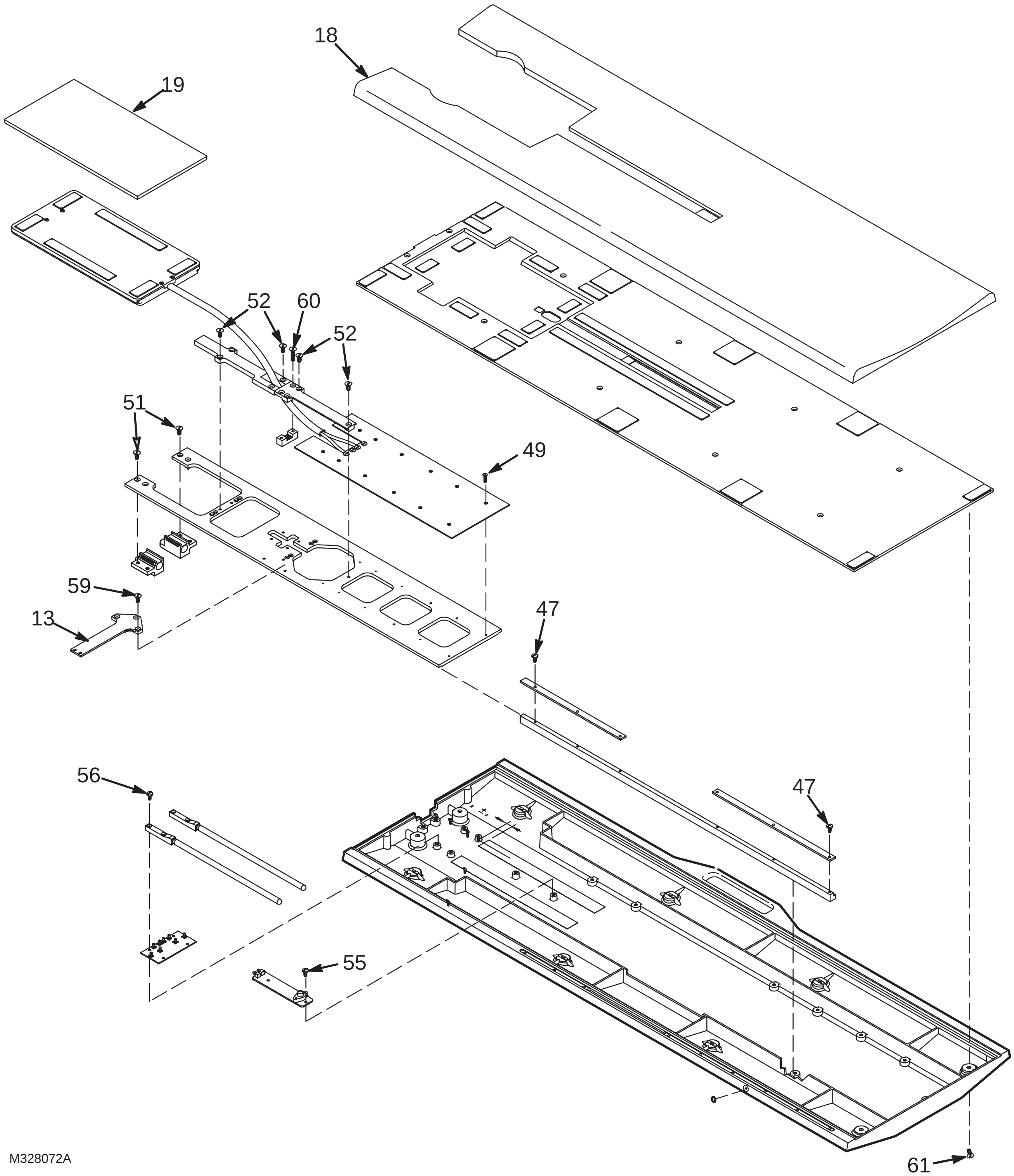
4-7 CLEANING THE PELVIC PHASED ARRAY COIL

Avoid damaging sensitive electronic parts. Do not spray or pour dishwashing solution directly onto the Pelvic Phased Array Coil, or external cable. Do not use alcohol to clean the Pelvic Phased Array Coil. Never submerge the Pelvic Phased Array Coil in any liquid.

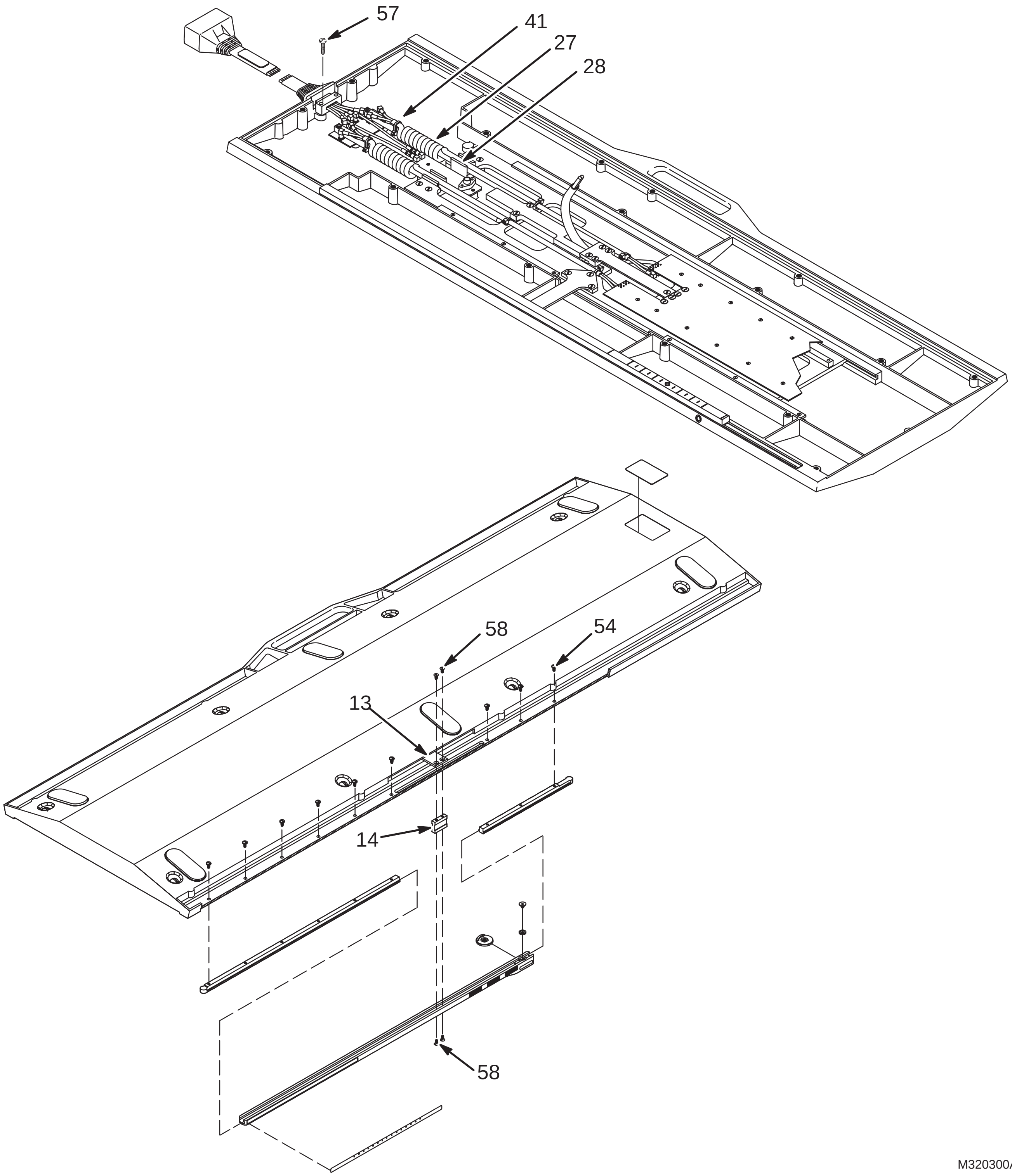
Clean the Pelvic Phased Array Coil and external cable with a mild dishwashing liquid and water solution. Wet a soft cloth with the solution and proceed to clean.

SECTION 5 – RENEWAL PARTS

PELVIC PHASED ARRAY COIL **46–320972G1** **(SEE PAGE 5–3 FOR PARTS LIST)**



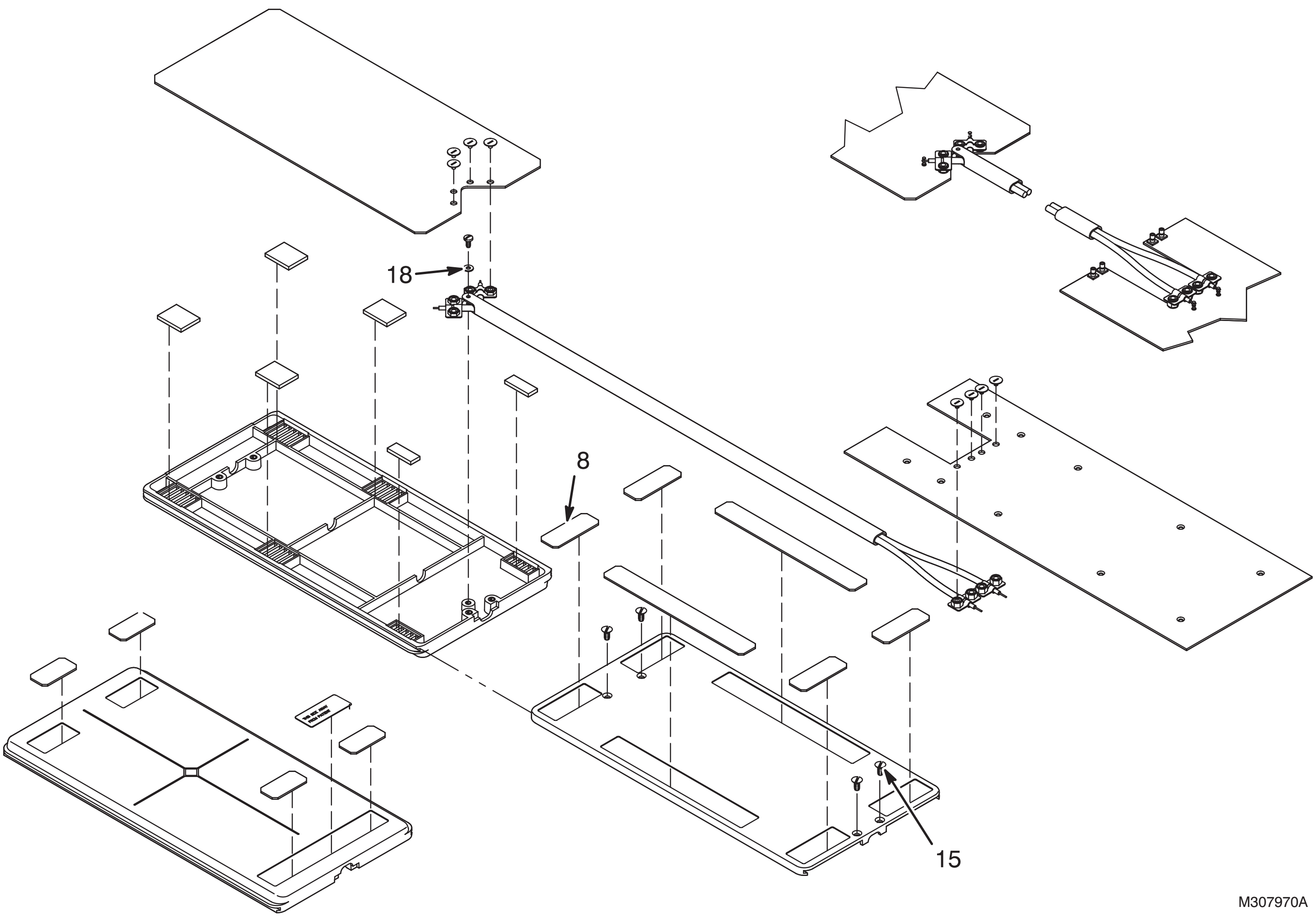
PELVIC PHASED ARRAY COIL 46–320972G1 (ILLUSTRATION CONT'D FROM PG 5–1)



PELVIC PHASED ARRAY COIL 46–320972G1 REV 11

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
13	46–307992P1	2	ARM	1	SLIDE PLATE ACTUATOR, DELRIN
14	46–307993P1	2	SPACER	1	.79"H X 1"W, 1/4" DELRIN, 6–32 HOLES
18	46–317449P1	2	COMFORT PAD	1	GREY NYLON FABRIC W/VELCRO HOOKS
19	46–317450P1	2	ENCL PAD	1	GREY NYLON FABRIC W/VELCRO HOOKS
27	46–317152P1	2	CABLE ASSEM	1	COILED CABLE, 2–COAX; 16–26, 15–25
28	46–317152P2	2	CABLE ASSEM	1	COILED CABLE, 2–COAX; 14–24, 13–23
31	46–317714P1	2	LABEL	1	"RAISE TABLE SIDE RAILS DURING EXAM"
41	46–208758P8	2	STRAP	6	5.50" X .140" SELF–LOCKING CABLE TIE
47	46–230007P5	2	SCREW,MACH	12	6–32 X 3/8" BIND HEAD SLOT, DELRIN
48	46–208921P22	2	SCREW,MACH	12	6–32 X 5/8", BRASS
49	46–230008P9	2	SCREW,MACH	10	4–40 X 5/8" LG FLAT HD, DELRIN
51	46–230008P1	2	SCREW,MACH	4	10–32 X 1/2", DELRIN
52	46–230008P11	2	SCREW,MACH	7	10–32 X 5/8" LG FLAT–HD, DELRIN
54	46–208921P28	2	SCREW,MACH	9	6–32 X 5/16" BINDING HEAD, BRASS
55	46–208921P4	2	SCREW,MACH	2	6–32 X 1/4", BRASS
56	46–208921P6	2	SCREW,MACH	4	6–32 X 1/2", BRASS
57	46–208921P10	2	SCREW,MACH	2	10–32 X 7/8", BRASS
58	46–208913P26	2	SCREW,MACH	4	6–32 X 3/8" LG, SLOT FLAT–HEAD BRASS
59	46–208913P12	2	SCREW,MACH	3	10–32 X 1/2", SLOT FLAT–HEAD, BRASS
60	46–208913P9	2	SCREW,MACH	2	10–32 X 3/4" LG, SLOT FLAT–HD, BRASS
61	46–220184P41	2	SCREW MACH	10	0010–32 X 0.500 LG RD HD SLOT BRASS

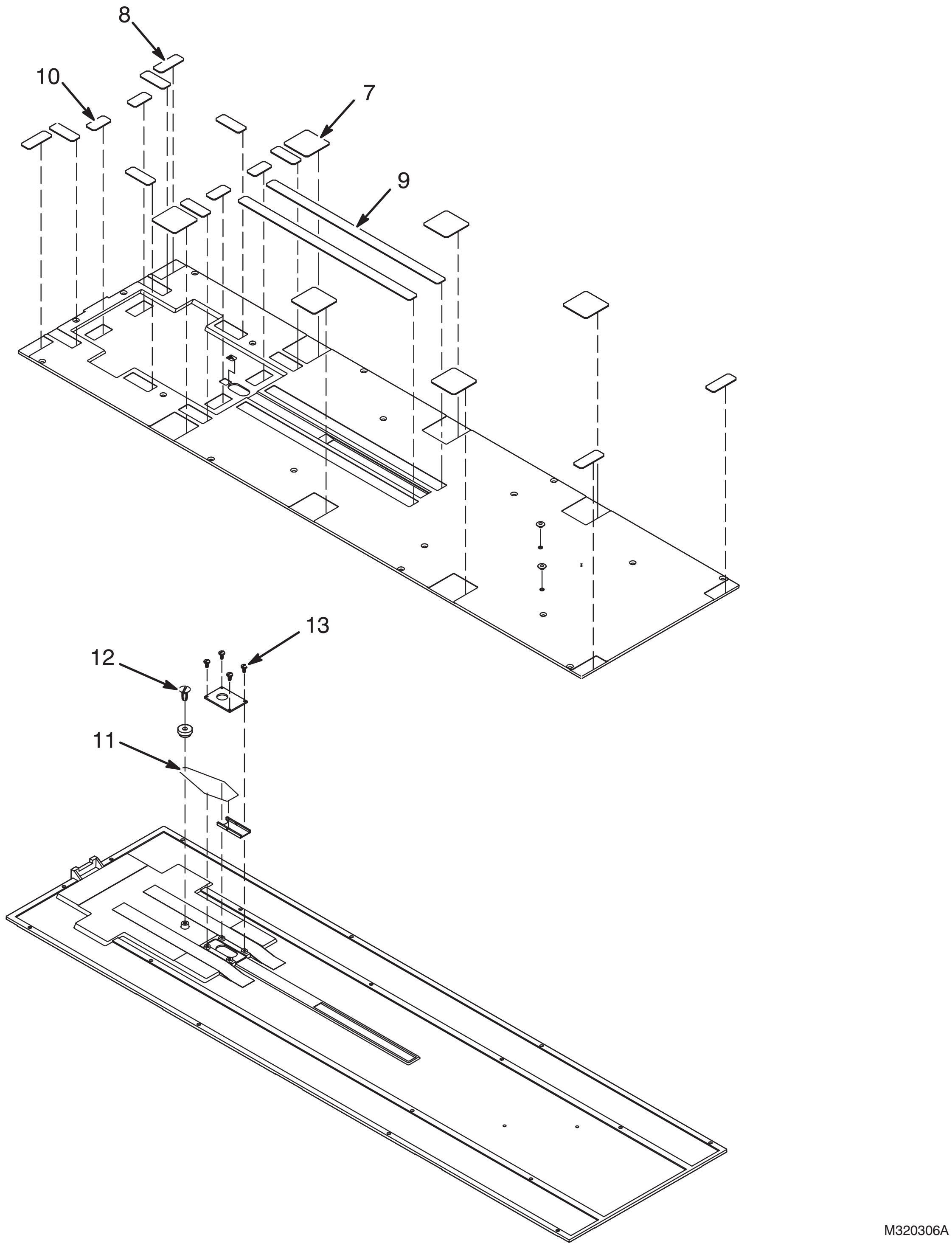
TOP M/C ARRAY ENCLOSURE ASSEMBLY 46–328084G1 REV 0



TOP M/C ARRAY ENCLOSURE ASSEMBLY 46–328084G1 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
8	46–317499P2	2	VELCRO LOOPS	4	3/4"W X 2"LG, ADHESIVE BACK, WHITE
15	46–230008P10	2	SCREW,MACH	4	6–32 X 0.375 LG FL HD DELRIN
18	1000904P333	2	WASHER,FLAT	1	ID 0.1285 OD 0.3125 THK .0313

COVER PLATE ASSEMBLY 46–328014G1 REV 0



COVER PLATE ASSEMBLY 46–328014G1 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
7	46–317475P1	2	VELCRO LOOPS	6	2"W X 2"L, WHITE, ADHESIVE BACKED
8	46–317499P2	2	VELCRO LOOPS	10	3/4"W X 2"LG, ADHESIVE BACK, WHITE
9	46–317499P4	2	VELCRO LOOPS	2	3/4"W X 14.5"L, ADHESIVE BACK, WHITE
10	46–317500P1	2	VELCRO HOOKS	4	3/4"W X 1.5"LG, ADHESIVE BACK, WHITE
11	46–282610P1	2	RUBBER BAND	AR	#33 SIZE, PURE RUBBER, 3–1/2 X 1/8"
12	46–230008P10	2	SCREW,MACH	1	6–32 X 0.375 LG FL HD DELRIN
13	46–208921P15	2	SCREW,MACH	4	2–56 X 0.250 LG BIND HD BRASS

DATA TABLE

Use the space provided below to record the calculated signal to noise ratio (SNR) data obtained from Section 3, FUNCTIONAL CHECKS. After recording the SNR data obtained during the initial Array installation, record subsequent SNR data in column 2 – 5 below and the date they were obtained in column number 1 as a periodic QA check. If the ratio of any of the coils found in column 6 is not greater than 85%, then there is a problem in the Multicoil system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value: ________________

Pelvic Phased Array Coil SNR Value:
Coil 1 _________ Coil 2 _________ Coil 3 _________ Coil 4 __________

Date: ____________________

SNR DATA QA (QUALITY ASSURANCE) CHECK:

1	2	3	4	5	6	
Date:	Coil 1 QA SNR Value:	Coil 2 QA SNR Value:	Coil 3 QA SNR Value:	Coil 4 QA SNR Value:	New values divided by original values above:	
						> 85% ?
						> 85% ?
						> 85% ?
						> 85% ?
						> 85% ?
						> 85% ?
						> 85% ?
						> 85% ?
						> 85% ?
						> 85% ?
						> 85% ?
						> 85% ?

Make additional copies of this document as is needed.

DEFECTIVE COIL RETURN FORM

Note

To allow for proper assessment of defective returned coils, this form must be completely filled out and accompany all returned coils. Include films or prints of any image quality related complaints with a description of scan protocol used.

Date____________________

Site Name____________________

Site Address____________________

Service Engineer____________________

Coil Serial Number____________________

Date Coil Installed____________________

Description of Coil Problem____________________

IMAGE QUALITY CHECKS

SNR Test
Body Coil SNR
Pelvic Phased Array Coil SNR

ELECTRICAL CHECKS

Pin Diode Test
Diode Drop Forward Bias
Diode Drop Reverse Bias

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" calibration procedure located in Direction 15491, *Signa Advantage 1.5T & 0.5T System Troubleshooting Guide*. Use Table DR–1 to record the TLT results of the defective coil.

TABLE DR–1
NC C TO NOTOI O UC ON C C I O

SITE:		NAME:	DATE:
TLT FILE NUMBER:			
SNR:	NOISE (0)		
	NOISE (1)		
	NOISE (2)		
	NOISE (3)		
	NOISE (C)		
	SNR MEAN (0)		
	SNR MEAN (1)		
	SNR MEAN (2)		
	SNR MEAN (3)		
	SNR MEAN (C)		
TR MAP:	89–91 (C)	%	%
(Combined Results)	85–95 (C)	%	%
	65–115 (C)	%	%
	FLIP MEAN (C)		
	FLIP SDV (C)		
	RCV MEAN (C)		
	RCV SDV (C)		
	DIFF MEAN (C)		
	DIFF SDV (C)		